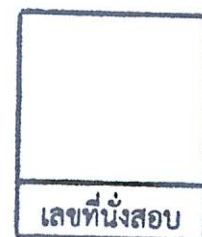


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Year: 2015

Subject: 392152 Chemistry II

Section: 15-16

Date: 10 March 2016

Time: 10.00-12.00

Name: _____ ID: _____ Field of Study: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Calculators are **NOT** allowed in the examination.
4. Dictionaries are **NOT** allowed
5. This is a closed book examination.
6. Electronic communication devices are **NOT** allowed in the examination room.
7. The examination has 6 pages (including this page), 2 sections and a total score of 55 points.
8. Write all your solution and answer on this examination sheet.

Relative atomic mass of

H = 1 C = 12 N = 14 O = 16 Na = 23 S = 32

Gas constant; $R = 0.082 \frac{\text{atm.L}}{\text{mol.K}}$ or $8.314 \frac{\text{J}}{\text{mol.K}}$

Part 1. Multiple Choice (19 points)

Choose the letter of the choice in the answer table that would be the best completes the statement or answers the question.

Question no.	a	b	c	d
1				
2				
3				
4				
5				
6				
7				
8				
9				

Question no.	a	b	c	D
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				

- This matter has a fixed shape and volume with particles closely packed together with little movement. It is _____.
a. solid b. liquid c. plasma d. gas
- Which one of the following states of matter can easily be compressed?
a. solid b. liquid c. gas and liquid d. gas
- Which one of the following is not a property of liquids?
a. No definite shape b. No definite volume c. They can flow d. They can diffuse
- Which of the following is NOT a way that matter changes phase?
a. sublimation b. melting c. mixing d. condensation
- Matter change in gas direct to solid is called _____.
a. sublimation b. melting c. mixing d. condensation
- An ideal gas is a hypothetical gas, _____.
a. not made of particles.
b. that conforms to all of the assumptions of the kinetic theory.
c. whose particles have zero mass.
d. made of motionless particles.

7. A shape of water droplet would be a good example for _____ ?



- a. viscosity b. surface tension
c. vapor pressure d. unit cell

8. At pressures greater than 1 atm, water will boil at a temperature _____.

- a. higher than 100°C. b. of 100°C. c. lower than 100°C. d. of 4°C.

9. Particles within a solid, _____.

- a. do not move. b. move about freely.
c. vibrate about fixed positions. d. exchange positions easily.

10. _____ is the amount of space that matter takes up.

- a. Pressure b. Volume c. Surface tension d. Unit cell

11. The vapor pressure of a substance increase as _____ increase

- a. viscosity b. molecular mass c. temperature d. surface area

12. Molality is defined as _____.

- a. kg solute/kg solution b. moles solute/ kg solvent
c. moles solute/ liter solution d. moles solute/mole solvents

13. Tyndall effect occurs only in _____.

- a. solution b. suspension c. colloid d. compound

14. Sedimentation occurs easily in _____.

- a. solution b. suspension c. colloid d. compound

15. Above sea level, water boils at a _____ temperature due to the _____ atmospheric pressure.

- a. higher, increase b. lower, increase c. higher, decrease d. lower, decrease

16. In which condition that the behavior of a real gas is similar to the behavior of an ideal gas?

- a. high temperature and high pressure b. high temperature and low pressure
c. low temperature and high pressure d. low temperature and low pressure

17. The rate of evaporation decrease as _____ increase.

- a. pressure b. temperature c. surface area d. flow rate of air

18. The pressure of sample of helium in a 1L container is 0.988 atm. What is the new pressure if the sample is placed in a 2L container?

- a. 0.245 atm b. 0.494 atm c. 1.00 atm d. 1.50 atm

19. Oxygen gas and sulfur dioxide (SO₂) gas are at the same temperature. What is the ratio of the effusion rates of their molecules?

- a. 0.15 b. 2 c. $\sqrt{0.15}$ d. $\sqrt{2}$

Part 2 Answer (36 points)

1. Fill out the blanks (3 points)

Changing	Changing of phase	Process	Endo/Exothermic
ice→water	solid→liquid	melting	endothermic
_____	liquid→gas	_____	_____
_____	_____	condensation	_____

2. The volume of gas at 100kPa is 300mL. If the pressure increases to 150kPa, what is the final volume? (2 points)

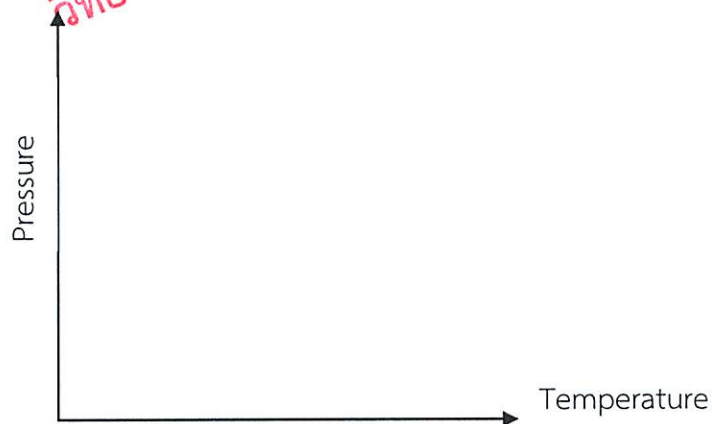
3. A gas at 27°C occupies a volume of 0.56L. At what temperature will the volume increase to 1.12L? (2 points)

4. A sample of gas occupies 2.0L at 227°C under 80.0 kPa. At what temperature would the gas occupy 200mL if the pressure increased to 100 kPa? (3 points)

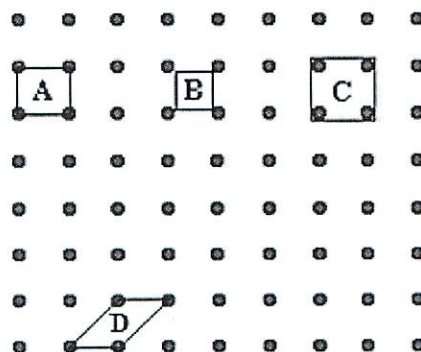
5. Find the density (as a function of R) of CO_2 at room conditions (25°C and 1 atm). (3 points)

6. Find the volume (as a function of R), in L, when 16.0 g of O_2 in a container with a pressure of 380 mmHg at a temperature of 27°C . (3 points)

7. Sketch the unary phase diagram (pressure vs temperature) of a pure substance. Label all phase fields. Indicate triple point and critical point. (3 points)



8. Which parallelograms in the figure on the right are unit cells? (2 points)



9. Complete the following table. (4 points)

	Concentration of a solution	Amount of the solute	Amount of the solution or solvent
9.1		0.02 g	10 mL solution
9.2	0.5 M.		6 L. solution
9.3		4.5 mol	3000 g solvent
9.4		4 mol	16 mol solution

10. Find the molarity of NaOH in a solution that contains 20 g of NaOH in 1500 mL solution?

(3 points)

11. A solution is prepared by dissolving 90 g of $C_6H_{12}O_6$ in 500 g of water to make 500 mL of solution. (8 points)

- Find the percentage concentration by mass (%w/w)
- Find the percentage concentration by mass per volume (%w/V)
- Find the molality of solution
- Find the mole fraction of $C_6H_{12}O_6$